MILITARY
SPECIFICATION

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Superseding
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ROTARY CONVERTERS (MOTOR GENERATORS) PU-134( )/U, PU-140( )/U, PU-141( )/U AND PU-143( )/U

1. SCOPE

1.1 Scope.- This specification covers the following rotary converters which are complete units of the fan-ventilated, drip-proof, semi-inclosed type equipped with radio-frequency filter components and installation accessories. (See 6.1 and 6.6)

Rotary Converter PU-134( )/U
Rotary Converter PU-140( )/U
Rotary Converter PU-141( )/U
Rotary Converter PU-143( )/U

2. APPLICABLE DOCUMENTS

2.1 Documents.- The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171	Bearings, Ball, Annular (General Purpose)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-571	Solder, Lead Alloy, Tin Lead Alloy, and Tin Alloy, Flux Cored Ribbon and Wire, and Solid Form
QQ-Z-325	Zinc Coating, Electrodeposited, Requirements for
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting

FSC 6125

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MIL-W-76	Wire and Cable, Hookup, Electrical, Insulated
MIL-P-79	Plastic Rods and Tubes, Thermosetting, Laminated
MIL-C-104	Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-C-132	Crate, Wood, Open, Maximum Capacity 2,500 Pounds
MIL-V-173	Varnish, Moisture-and-Fungus-Resistant
MIL-P-997	Plastic-Material, Laminated, Thermosetting, Electrical-Insulating, Sheets, Glass Cloth, Silicone Resin
MIL-V-1137	Varnish, Electrical-Insulating (for Electro-Motive Equipment)
MIL-M-2130	Motor-Generators, General Purpose (Naval Shipboard Use)
MIL-P-3115	Plastic Sheet, Laminated, Thermosetting, Paper-Base, Phenolic-Resin
MIL-G-3278	Grease, Aircraft and Instrument (for Low and High Temperatures)
MIL-C-3767	Connector, Receptacle, Electrical (Power, Two-Wire, Polarized, Twist-Locking Type), Type UR 101F
MIL-C-3774	Crates, Open, Wood (2,500 to 10,000 Lb)
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-I-11748	Interference Reduction for Elect Equip
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base)
MIL-P-15047	Plastic-Material, Laminated Thermosetting Sheets, Nylon Fabric Base, Phenolic-Resin

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-169	Extreme-Temperature Cycle
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-705	Generator Sets, Engine-Driven Methods of Tests and Instructions

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DRAWINGS

SIGNAL CORPS

SC-A-46439	List of Accessories for Package Tester Motor-Generator PU-143( )/U
SC-DL-328000	
SC-A-362100	Requirements for Preparation of Packaging and Packing Illustrations and Bills of Material

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Description.- The Rotary Converters PU-134( )/U, PU-140( )/U, PU-141( )/U and PU-143( )/U are used to convert various dc voltages (See 3.7.2) to 110 volt, 60 cycle per second (cps), ac, at a power factor of 85 percent. These units are equipped with radio-frequency filter components, inspection covers and mounting feet.

3.2 Preproduction samples.- The contractor shall furnish two (2) preproduction samples of each type of rotary converter on order for approval, provided the invitation for bids and the contract requires samples. (See 4.3)

3.3 Construction.-

3.3.1 Rotary Converter, PU-143( )/U.- Rotary Converter PU-143( )/U shall be constructed in accordance with Drawing and Data List SC-DL-328000.

3.4 Physical limits.-

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3.4.1 Dimensions.- The overall dimensions of Rotary Converters PU-134( )/U, PU-140( )/U and PU-141( )/U shall not exceed 14 inches in length, 8 inches in width and 14 inches in height. The dimensions of the filter box and shock mounts are included in this determination.

3.4.2 Weight.- The total weight of Rotary Converters PU-134( )/U, PU-140( )/U and PU-141( )/U shall not exceed 70 pounds.

3.5 Parts and materials for Rotary Converters PU-134( )/U, PU-140( )/U and PU-141( )/U.-

3.5.1 Parts, materials, and processes; general.- Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements, as specified herein. Ratings and derated values of parts and materials shall not be exceeded when the equipment is subjected to specified service conditions. (See 3.10) Derating methods used shall be as specified by the applicable individual documents covering the parts and materials.

3.5.2 Field.- The field shall be of the compound wound stationary type. Interpole windings may be provided, if required, to permit compliance with the regulation requirements of 3.7.3.

3.5.3 Armature.- The armature shall be of the rotating type and shall consist of two sets of windings; the d-c motor windings connected to a commutator, and the a-c windings connected to collector rings. The rotor shall be mechanically balanced in accordance with paragraph 3.1.10, and tested per 4.5.20 per Specification MIL-M-2130.

3.5.4 Collector rings.- Collector rings shall be made of brass, bronze, or other corrosion-resistant material of suitable composition for the application. They shall be smooth and free from cuts, flaws, or other irregularities.

3.5.5 Commutator.- The commutator segments shall be of the best quality hard drawn copper and shall be insulated from the shaft and each other by a high grade mica plate, pressed mica plate, or other approved material. If the insulation between bars is under-cut, all sharp edges shall be removed from the bars. The commutator shall be rigidly secured to the shaft by keying or other approved method which will insure maintenance of alignment with respect to the armature core slots. The commutator shall be thoroughly tightened and baked prior to testing the machine. Commutator surfaces shall run true and shall be free from flaws, tool marks, or other imperfections.

3.5.6 Brush holders.- All brush holders shall be readily accessible. They shall be of corrosion-resistant material designed to prevent binding of brushes, and to permit easy removal of brushes for inspection and replacement. Cadmium plating will not be permitted on brush holders. Springs shall be of phosphor bronze or suitably protected steel and shall not be used to carry current. Brush holders shall be separately adjustable for spring tension and properly arranged to insure minimum uniform wear of commutator or collector rings, both through mechanical abrasion and electrolytic action. Brush holders shall be finished smooth, permit free movement of the brushes and shall be designed in such a manner that the point of contact between the brushes and commutator or collector rings will not shift by the wearing away of the brushes. Brush holders shall be mounted so that brushes are suitable under all operating conditions and so that all clearances are uniform.

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3.5.7 Brushes.- All brushes shall be of a high grade homogeneous material properly selected for the particular application. The characteristics of the brushes shall be such that they will give satisfactory operation without chattering, cutting, or burning the rings or commutator and such that abnormal wear will not occur at low temperatures. Flexible brush shunts shall be provided for carrying current and shall be firmly fixed to the brush and brush holder in such a manner that the connections will not loosen under vibration or heat. Suitable protection shall be provided to prevent grounding of brush shunts. Each brush shall be formed to the contour of the commutator or collector ring for which it is intended. All brushes shall be standard in design, interchangeable in sets and commercially available.

3.5.8 Filter box.- The filter and suppression elements shall be housed in a drip-proof metal box (See 4.14) that shall be securely mounted on top of the rotary converter. Input leads shall be brought out from the filter box through holes which are equipped with rubber bushings.

3.5.9 Bearings.- The armature shall be supported by double sealed ball bearings which are in accordance with Specification FF-B-171 and packed with grease which conforms to the requirements of Specification MIL-G-3278.

3.5.10 Frame.- The frame shall be of high-grade, non-fragile cast iron, cast steel or rolled steel. It shall be carefully machined and shall permit easy removal of the rotor.

3.5.11 Mounting supports.- The converter shall be provided with four mounting pads suitable for base mounting on standard shock-absorbing vibration isolators.

3.5.12 Power outlet.- A two wire polarized, twist locking type, power, electrical receptacle connector type UR101F per Specification MIL-C-3767 shall be provided.

3.5.13 Hook-up wire.- Hook-up wire used in wiring control and suppression circuits shall be in accordance with Specification MIL-W-76 and shall be stranded wherever practicable. The cross-sectional area of the conductor shall be sufficient to carry the necessary current with a minimum of overheating, such that the service life of the equipment will not be shortened by electrical or mechanical deterioration of the wire from that cause.

3.5.14 Insulation material.- Insulation material shall not support rapid combustion. The material used shall be sufficiently rigid and free from cold flow to maintain its function during exposure of the equipment in specified operating conditions and to the tests specified in Section 4.

3.5.15 Thermosetting plastic.- Unless otherwise specified, plastic material and parts shall conform to Table I and shall be suitable for the intended application. The material shall be treated in accordance with Table II.

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Table I.- Thermosetting plastic material and parts

Description	Specifications	Types
Laminated rods and tubes	*MIL-P-79	All types
Laminated sheets, glass-cloth, silicone-resin	MIL-P-997	GSG
Laminated sheets, nylon fabric base, phenolic-resin	MIL-P-15047	NPG
Laminated sheets, paper-base, phenolic-resin	MIL-P-3115	PBE, PBG
Molding material and molded parts	MIL-M-14	All types

* Fabric base laminates shall not be used unless otherwise specified or unless required for mechanical applications only.

Table II.- Treatment of thermosetting plastic material

Material	Treatment
CMG, CMI-5, CMI-10, MAG, MAI-30, MAI-60, MDG, MME, MMI-5, MMI-30, MSG, MSI-30, SDG, SDG-F, SDI-30 per Specification MIL-M-14 GMG per Specification MIL-P-79 GSG per Specification MIL-P-997 NPG per Specification MIL-P-15047	Do not treat; mask, if necessary during treatment of other items.
GPI-100, MFE, SDI-5 per Specification MIL-M-14 PBE per Specification MIL-P-79 PBE, PBE-P per Specification MIL-P-3115	Treatment not required; need not be masked during treatment of other items.
All unmachined, molded materials except those listed above	Treat with varnish conforming to Specification MIL-V-173
All laminated materials and all machined, molded materials, except those listed above.	Treat all surfaces with insulating varnish, type M, grade CB, conforming to Specification MIL-V 1137, and bake for the time and at the temperature specified by the manufacturer of the varnish.

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3.5.16 Screws, other threaded devices, and related parts.-

3.5.16.1 Form and fit.- Unless otherwise specified, screw threads shall conform to Handbook H28; the Unified Thread Form and Series shall be class 2A or 2B fit, and the American National Thread Series shall be class 2 fit.

3.5.16.2 Material.- Where permitted by design considerations, and unless otherwise specified, screws, other threaded devices, and related parts shall be of carbon steel having cadmium plating, type II, class 3, conforming to Specification QQ-P-416; of steel having zinc plating, type II, class 3, conforming to Specification QQ-Z-325; or of corrosion resistant steel, passivated.

3.5.17 Insulation and impregnation.- The converter windings shall be impregnated, preferably by the "vacuum" process, with a minimum of two coats of insulating varnish, Grade CB, in accordance with Specification MIL-V-1137. A third or sealer coat, selected for compatability with the varnish used and resistant to moisture, acids, petroleum products, etc., shall then be applied. Drying and baking schedules shall be in accordance with the recommendations of the manufacturers of the materials used. This treatment shall insure uniform penetration throughout the windings.

3.6 Process requirements for rotary converters PU-134( )/U, PU-140( )/U, PU-141( )/U and PU-143( )/U.-

3.6.1 Plastic material and parts.- Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.6.2 Riveting.- Riveting shall not be used for mounting parts such as capacitors, resistors, and inductors. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1-1/2 times the rivet diameter. Rivets shall be used where practicable in preference to other means of securing aluminum or magnesium structural parts if the joint is subjected to dynamic tension loads when the equipment is exposed to specified service conditions and if the parts do not require removal when the equipment is serviced. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.6.3 Soldering.-

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3.6.3.1 Solder.- Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition SN60, type AR or S, shall be used for general purposes; and composition SN62 or SN63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.6.3.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.6.3.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.6.4 Wiring.- Wiring shall be neat and sturdy.

3.6.4.1 Slack.- Wires shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on wires and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

3.6.4.2 Protection.- Wires shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal partitions, shields, or similar items less than 1/8 inch in thickness, the wires shall be protected by suitable grommets or bushings. Where wires run through holes in metal 1/8 inch in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1/16 inch.

3.6.4.3 Support.- Wire shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire (1) during and after subjection of the equipment to specified service conditions or (11) after service or repair of the equipment in a normal manner. Twine or tape shall not be used for securing wire.

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3.6.4.4 Clearance.- Clearance between wires and parts, such as resistors shall be provided to avoid deterioration of the wires because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.6.4.5 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.6.5 Finish, protective.- Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3)

3.6.6 Marking.- Marking shall conform to Specification MIL-M-13231. (See 4.4)

3.6.6.1 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.6.2 Serial numbers.- Each rotary converter on order shall be serial numbered.

3.6.7 Tropicalization of material.- Parts of the filter circuit fabricated of thermosetting plastic containing cotton, linen or wood-flour as a filler or base shall be treated in accordance with paragraph 3.6.7.1 through 3.6.7.4 and paragraph 3.5.15.

3.6.7.1 Treating materials.-. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.6.7.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.6.7.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

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3.6.7.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2(j))

3.6.8 Securing of parts.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.6.9 Cleaning.-

3.6.9.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.6.9.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.7 Electrical characteristics of rotary converters PU-134( )/U, PU-140( )/U, PU-141( )/U, and PU-143( )/U.-

3.7.1 Rating.- Each rotary converter shall be rated at 500 volt-amperes at 85 percent lagging power factor, 110 volts, single phase, 2 wires, 60 cycle per second, at 3600 rpm, with a temperature rise not to exceed 40°C.

3.7.2 Input.- The input values of the rotary converters shall be as specified in Table III below:

Table III.- Input characteristics

Rotary Converter	Voltage, dc	Amperes, dc
PU-134( )/U	230	4.1
PU-140( )/U	115	6.1
PU-141( )/U	32	18.0
PU-143( )/U	24	27.0

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3.7.3 Regulation.- When tested under any of the conditions listed in 3.10.1 and 3.10.2 the voltage and frequency of the output of each rotary converter shall be within the limits specified in Table IV.

Table IV.- Output characteristics

Input volts for each rotary converter (See Table III)	Output - No Load		Output-Full Load at 85% P.F.	
	Volts	*Frequency c.p.s	Volts	*Frequency c.p.s
Rated input	125	63.0	110	60.0
Rated input plus 10%	139	68.0	118	64.0
Rated input minus 10%	113	58.0	94	54.0

*A plus or minus two percent (±2%) tolerance is permitted.

3.7.3.1 Adjustment.- Suitable means shall be provided to enable adjustment of the rotary converter so that the output will be within the limits specified in 3.7.3 for any input voltage from 10 percent above to 10 percent below the rated value.

3.7.3.2 Frequency regulation.- At any single ambient temperature and at a constant input voltage, the frequency regulation between no-load and full load at 85% power factor shall not exceed 2 cps.

3.7.4 Wave form.- The value of any harmonic shall not exceed 5 percent of the fundamental. The change in crest factor from no load to full load at any input voltage shall not exceed 10 percent.

3.8 Commutation.- Commutation for rotary converters PU-134( )/U, PU-140( )/U, PU-141( )/U and PU-143( )/U, shall be such that no harmful sparking occurs. The brushes shall be straight in the holders and shall be so seated that at least 75 percent of the bearing surface of each brush is in contact with the commutator.

3.9 Suppression of radio interference.- Rotary Converters PU-134( )/U, PU-140( )/U, PU-141( )/U and PU-143( )/U shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for class 1. Application of suppression components and techniques shall conform to that specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits, he shall obtain approval from the contracting officer before proceeding with the modification.

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3.10 Service conditions for rotary converters PU-134( )/U, PU-140( )/U, PU-141( )/U, and PU-143( )/U.- The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.10.1 Temperature (see 4.17).-

(a) Operating: Ambient temperature in the range of +150°F to -18°F. (The 150°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.10.2 Relative humidity (See 4.13).- Daily exposure of up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.10.2.1 Insulation resistance after moisture resistance test.- Following subjection to the moisture resistance test, specified in 4.13, the insulation resistance shall not be less than 50,000 ohms (See 4.15).

3.10.3 Bounce.- After being subjected to the test of paragraph 4.21, the equipment shall show no physical damage, except for minor surface abrasion, and shall be capable of meeting the requirements of this specification.

3.10.4 Vibration, internal.- The amplitude of vibration of any part, sub-assembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second.(See 4.20).

3.10.5 Performance.- The rotary converter shall be capable of being operated at rated output for 500 hours. At the completion of the test, the unit shall meet the requirements of 3.7.3. Evidence of excessive wear or deterioration of any component, resulting from this test, shall be considered cause for rejection.

3.10.6 Heat run.- The sample unit shall be capable of being operated at rated output for a period of two hours in an ambient temperature of 50°C. The temperature of any winding, when measured by the resistance method, shall not exceed 90°C. There shall be no overheating of parts during this test and the units shall show no excessive vibration, noise or erratic regulation. (See 4.9)

3.10.7 Operational.- The equipment shall be capable of meeting the test of paragraph 4.16 to indicate that it is operable.

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3.10.8 High potential.- Electrical insulation of the equipment shall be capable of withstanding a minimum breakdown voltage of 1500 volts dc. After a 1 minute application of the voltage, there shall be no visible deterioration and the equipment shall meet the specified performance requirements. (See 4.11)

3.11 Interchangeability applying to rotary converters PU-134( )/U, PU-140( )/U, PU-141( )/U and PU-143( )/U.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.12) Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.12 Preconditioning applying to rotary converters PU-134( )/U, PU-140( )/U, PU-141( )/U and PU-143( )/U.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.19. (Also see 4.5)

3.13 Technical literature and running spare parts.- Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.14 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.6.1 Plastic material and parts
3.6.2 Riveting
3.6.3.2 Acid or acid salts
3.6.3.3 Soldering process
3.6.4 Wiring
3.6.4.1 Slack in wires
3.6.4.2 Protection of wires
3.6.4.4 Clearance in wiring
3.6.4.5 Connections
3.6.8 Securing of parts
3.6.9 Cleaning

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the exmaination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5)

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(2) Acceptance inspection of preparation for delivery. (See 4.22)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified by the contract. It shall consist of the preproduction inspection specified in Table V, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A (See Table VI), group B, and group C (See Table VII). The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce and (3) performance; other preproduction inspection may precede, follow or be interspersed between the foregoing.

Table V.- Preproduction Inspection

Inspection (For additional preproduction inspection see 4.3)	Req. Para.	Insp. Para.
Wave form	3.7.4	4.8
Humidity test	3.10.2	4.13
Drip proof	3.5.8	4.14
Insulation resistance after moisture resistance test	3.10.2.1	4.15
Performance inspection	3.10.5	4.7
Temperature test	3.10.1	4.17
Radio frequency interference test	3.9	4.18

4.4 Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery.

Item	Req. Par.
Finish	3.6.5
Marking	3.6.6

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test equipment for Government verification inspection shall be made

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available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.12)

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to Table VI, and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.16) shall be done last. (See 6.5)

Table VI.- Group A inspection

Inspection	Req. Para	Insp Para	AQL major	AQL minor
Visual and mechanical	3.14	4.6	1.0%	4.0%
Electrical				
Heat run	3.10.6	4.9	1.0% for	
Voltage and frequency regulation	3.7.3	4.10	the group	
High potential test	3.10.8	4.11	combined	
Operational test	3.10.7	4.16	1.0%	

4.5.2 Group B inspection.- This inspection shall consist of the test specified for interchangeability in paragraph 4.12. The inspection, including sampling, shall conform to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.3 Group C inspection.- This inspection shall be as listed in Table VII, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table VII.- Group C Inspection

Inspection	Req. Para.	Insp. Para.
Vibration, internal	3.10.4	4.20
Bounce	3.10.3	4.21

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4.5.3.1 Sampling for inspection of equipment.- One rotary converter out of each 50 or fraction thereof for group C inspection shall be selected each month, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4)

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252 and all other defects thay may reduce the efficiency of the item or make it inoperable or hazardous

4.7 Performance inspection.- The equipment shall be tested to determine compliance with the requirement specified in paragraph 3.10.5.

4.8 Wave form.- The crest factor and harmonic content of the output voltage shall be determined under the following conditions.

(a) At minimum voltage for no load, half load, and full load (1.0-power-factor-steady-resistive load).

(b) Test (a) shall be repeated for steady-minimum-lagging power factor.

(c) Tests (a) and (b) for rated input voltage shall be repeated.

(d) Tests (a) and (b) for maximum input voltage shall be repeated.

The crest factor obtained by the above test shall be 1.414 ±10 percent. The crest factor and value of any harmonic shall be as specified in paragraph 3.7.4.

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The crest factor shall be determined by means of an oscillograph or peak and rms voltmeters. When voltmeters are used, the peak reading instrument shall be connected successively for each of the two possible polarities, and the crest factor shall be the ratio of the average of the two peak values of the reading of the rms voltmeters.

The harmonic content shall be determined by an approved analyzer.

4.9 Heat run.- During this test, the following data shall be observed and recorded: ambient temperature, frame "hot spot" temperature (by thermometer), input voltage and current, output voltage and current, frequency, and temperature of windings and bearings. The equipment shall also be inspected for excessive vibration, noise and erratic regulation. Units that do not meet the requirements of paragraph 3.10.6 shall be rejected.

4.10 Voltage and frequency regulation.- At the conclusion of the heat run, the convertor shall be checked at each of the following points: o, 1/2, 3/4 and full rated load for compliance with the requirements of paragraph 3.7.3.

4.11 High potential test.- The equipment shall be subjected to a high potential test in accordance with Method 302.1 of Standard MIL-STD-705 to determine compliance with the requirement specified in paragraph 3.10.8.

4.12 Inspection for interchangeability.- The equipment shall be examined for compliance with the requirements of paragraph 3.11. The following measurements are to be included in the examination and when one of these dimensions is not within specified or design limits, it shall be considered a major defect.

(a) The size and location of bolt and screw holes in mounting feet, inspection covers, end brackets, baffle plate, filter boxes, filter box covers etc.

(b) The internal and external dimensions of filter boxes and filter box covers.

(c) The inside diameter, outside diameter and depth of end brackets and field yokes.

4.13 Moisture-resistance test.- The equipment shall be tested as follows:

4.13.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

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4.13.2 Test method.-

(a) Dry at 130° ± 5°F for 24 hours.
(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.
(c) Take measurements as specified in 4.13.3, and readjust or realine as necessary to meet full specification requirements.
(d) Subject to continuous cycling for 5 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.13.3 shall be made during the periods shown on the standard and shall comply with 4.13.3.
(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.13.3.

4.13.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.7	Performance inspection	Full specification requirements
4.10	Voltage and frequency regulation	Full specification requirements
4.15	Insulation resistance	Full specification requirements

4.13.4 Failure.- If any equipment fails to meet the performance specified in 4.13.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.13.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.13.2(e), it does not pass the test.

4.14 Drip proof.- Immediately upon removal of the equipment from the humidity chamber, the filter box shall be opened and inspected. Visible evidence of leakage will constitute noncompliance with the drip proof requirement of paragraph 3.5.8

4.15 Insulation resistance.- The equipment shall be subjected to the insulation resistance test specified in Method 301.1 of Standard MIL-STD-705 to determine compliance with the requirement of paragraph 3.10.2.1.

4.16 Operational test.- The rotary converters shall be connected to a d.c. source and the output voltage checked by a voltmeter. A reading on the voltmeter shall constitute compliance with the requirement. (See 3.10.7)

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4.17 Temperature test.- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169 except that step 8 shall be at -18°F. Performance shall be as follows:

Step	Paragraph	Measurement	Performance
1,3,8, & 10	4.7	Performance inspection	Full specification requirements

4.18 Radio frequency interference test.- A preproduction sample of each rotary converter shall be tested to determine compliance with the requirement specified in paragraph 3.9.

4.19 Bounce preconditioning.- The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.20 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.10.4:

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.21 Bounce test: class F units.- The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

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(b) Place the equipment prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.22 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.23 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (See 6.2(e)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Paragraph	Measurement
4.7	Performance inspection

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- Each rotary converter shall be preserved, packaged and tested in accordance with the applicable provisions of Specification MIL-P-116 and in a manner that will afford adequate protection against corrosion, deterioration and damage during world-wide shipment, handling and open storage.

5.1.2 Level C.- Each rotary converter shall be afforded preservation and packaging in accordance with the suppliers' normal commercial practice.

5.2 Packing.- Each rotary converter packaged as specified, shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level. The gross weight shall not exceed the weight shown unless an individually packed item exceeds that amount. Closure and strapping shall be as prescribed in the applicable box specification or appendix thereto, except that bands of reinforced, pressure-sensitive adhesive tape, 1/2 inch wide and conforming to Specification PPP-T-97 for Grade IV, shall be used for fiberboard boxes in lieu of metal strapping.

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5.2.1 Level A.-

BOX SPECIFICATION		MAX. GROSS WT.
Fiberboard (Type I, Class 2)	PPP-B-636	Table VI
Nailed Wood (Style 4, Class 2)	PPP-B-621	200 lbs.
Wirebound (Style 2 or 3, Class 2)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Overseas)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Overseas)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.
Crate, Wood, Sheathed	MIL-C-104	Specification Limit
Crate, Wood, Open	MIL-C-3774	"
Crate, Wood, Open	MIL-C-132	"

5.2.2 Level B.-

BOX SPECIFICATION		MAX. GROSS WT.
Fiberboard (Type I, Class 1)	PPP-B-636	Table I
Nailed Wood (Style 4, Class 1)	PPP-B-621	200 lbs.
Wirebound (Style 3, Class 1)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Domestic)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Domestic)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.
Crate, Wood, Open	MIL-C-3774	Specification Limit
Crate, Wood, Open	MIL-C-132	"

5.2.3 Level C.- Each rotary converter shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.2.4 Pilot Pack.- When Level "A" packaging and packing is specified above, one acceptable model of each rotary converter will be pilot packed in a manner that will pass the preproduction tests prescribed in Specification MIL-P-116. Illustrations of the packaging and packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing No. SC-A-362100A.

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5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- The rotary converters are used to convert various dc voltages (See 3.7.2) to 110 volt, 60 cycle per second, ac, at a power factor of 85 percent. Their intended use is to supply ac power for fixed station and shipborne radio receiving and transmitting equipment.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.
(b) Type of rotary converter required.
(c) Level of packaging and level of packing required for shipment. (Level A, level B or level C)
(d) The specific paragraphs of Section 5 which are applicable to the particulat procurement.
(e) Preproduction pack(s) as follows:
Makeup of pack(s)
Number of each kind of pack to be submitted
Inspection to be performed thereon
(f) Marking and shipping of samples.
(g) Place of final inspection.
(h) Technical literature required. (See 3.13)
(i) Quantity of running spare parts required. (See 3.13)
(j) Submission of the statement of treatment referenced in 3.6.7.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Color.- The color chip furnished by the procuring agency (3.6.5) will be color chip. No. X-24087, and may be obtained upon request to Commanding General, U.S. Army Signal Supply Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania ATTN: SELMA-J4b.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.5 Location of operational inspection.- It is desirable that the operational inspection (4.16) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PU-134W/U. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.7 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A--The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B--That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C--A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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